

A loss-based framework for coherent prediction of competing risks at prespecified time horizons

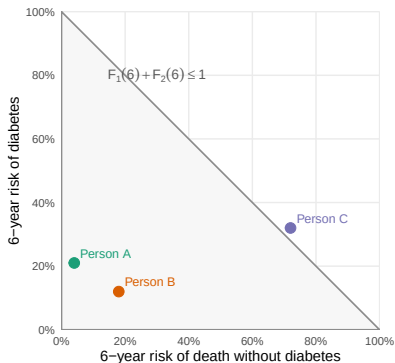
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Predicting the absolute 6-year risk of diabetes in presence of the competing risk of death

$$F_1(6 \mid x) = P(\text{diabetes by 6 years} \mid x),$$

$$F_2(6 \mid x) = P(\text{death without diabetes by 6 years} \mid x).$$



Prediction modeling strategy

Learner library $\longrightarrow$ K-fold cross-validation $\longrightarrow$ Super learner ensemble $\longrightarrow$ External validation

Candidate learners include cause-specific Cox models, IPCW binomial regression and machine learning methods suitably adapted for competing risk outcomes.

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Focus of this talk

- Parametric Cox regression models for both causes estimated coherently by minimizing the prediction time horizon-specific Brier-score. A learner that ensures $F_1(6|x) + F_2(6|x) \leq 1$.
- If and when to artificially censor subjects at the prediction time horizon.

Setting: competing risks

Notation

- T : the time between time zero¹ and the date of an event
- $D \in \{1, 2\}$: the cause of the event
- $X = (X^1, \dots, X^p)$: a p -dimensional vector of baseline covariates
- C : the time between time zero and censoring
- $\tilde{T} = \min(T, C)$: observed time
- $\Delta = 1\{T \leq C\}$: censoring indicator
- $\tilde{D} = \Delta D$: observed event
- τ : prediction time horizon

We assume conditional independent censoring and that the probability of being censored at time τ is bounded away from zero for all x .

¹Any time in the life of a person

The parametric Cox-exponential model

For both causes $j \in \{1, 2\}$ and censoring $j = 0$, specify Cox regression models with constant baseline hazards:

$$\lambda_j(t | x) = \lambda_{0j} \exp(x^\top \beta_j), \quad \lambda_{0j} > 0,$$
$$\Lambda_j(t|x) = \int_0^t \lambda_j(s|x) ds = t \lambda_j(t|x)$$

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The observed cause-specific cumulative incidence functions:

$$\begin{aligned}\tilde{F}_j(t|x) &= P(\tilde{T} \leq t, \tilde{D} = j|x) \\ &= \int_0^t \exp\{-\Lambda_1(s - |x) - \Lambda_2(s - |x) - \Lambda_0(s - |x)\} \Lambda_j(ds|x)\end{aligned}$$

Estimation of observed risks ²

Brier-score loss function

$$BSE = \frac{1}{N} \sum_{i=1}^N \sum_{j=0}^2 \{ \tilde{F}_j(\tau \mid X_i) - 1(\tilde{D}_i = j, \tilde{T}_i \leq \tau) \}^2.$$

²Munch & Gerds (2026). The joint survival super learner: A super learner for right-censored data. *Statistica Neerlandica*, 80(2), e70026.

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- Estimating the parameters of the three Cox-exponential models by minimizing the overall Brier score yields $\hat{\Lambda}_1, \hat{\Lambda}_2, \hat{\Lambda}_0$
- No need for IPC weights
- Derive estimates of the absolute risks of both events $\hat{F}_1, \hat{F}_2$ based on $\hat{\Lambda}_1, \hat{\Lambda}_2$

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Brier score estimation

Parameter vector and the estimator:

$$\theta = (\lambda_{00}, \beta_0, \lambda_{01}, \beta_1, \lambda_{02}, \beta_2),$$
$$\hat{\theta}_{BSE} = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \sum_{j=0}^2 \left\{ \tilde{F}_j(\tau|x; \theta) - 1(\tilde{D}_i = j, \tilde{T}_i \leq \tau) \right\}^2.$$

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Cause specific risk predictions can then be obtained by substituting $\hat{\theta}_{BSE}$ into:

$$\begin{aligned} F_j(t|x) &= P(T \leq t, D = j|x) \\ &= \int_0^t \exp\{-\Lambda_1(s - |x; \theta) - \Lambda_2(s - |x; \theta)\} \Lambda_j(ds|x; \theta) \end{aligned}$$

Why estimate like this?

Working hypothesis:

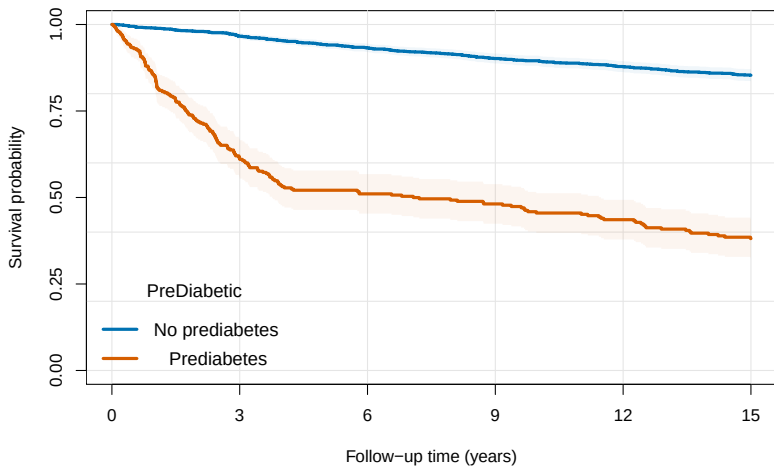
The Cox-exponential model makes sense as we only need predictions for one specific timepoint, so we don't need the baseline hazard to vary over time.

Estimating both cause-specific hazards this way:

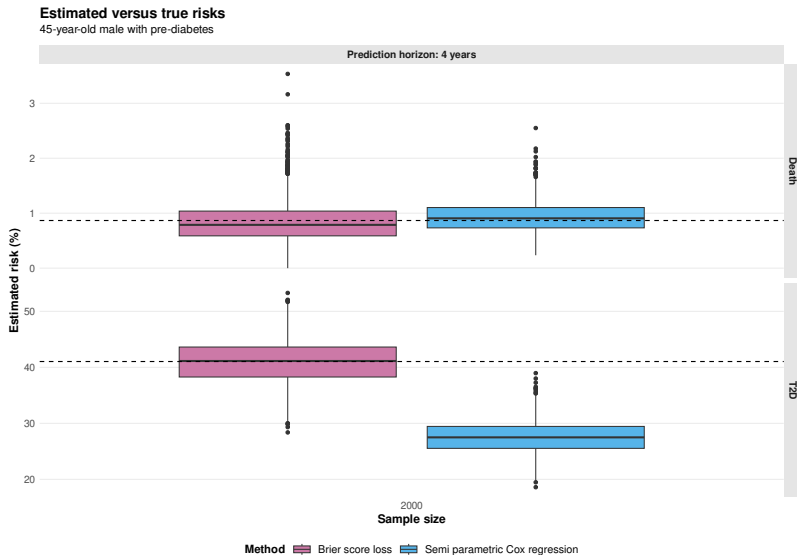
- targets calibration and discrimination directly at the prediction horizon
- performs well in non-proportional hazards data
- respects the parameter space, i.e., $F_1 + F_2 \leq 1$
- produces a portable parametric learner for use outside protected environments.

Non-proportional hazards data

Diabetes-free survival curves by prediabetes status



Simulation study: Prediction at 4 years



Artificial censoring at the prediction horizon

Artificial censoring replaces the observed outcome by

$$\tilde{T}_i^{(4)} = \min(\tilde{T}_i, 4), \quad \tilde{D}_i^{(4)} = \tilde{D}_i 1\{\tilde{T}_i \leq 4\}.$$

It removes all event-time information after the four-year horizon. This affects the estimates of Cox models fitted using likelihood methods.

It can lose predictive power

If proportional hazards is correct, later events inform the same β_j that governs hazards before four years. Discarding them wastes information.

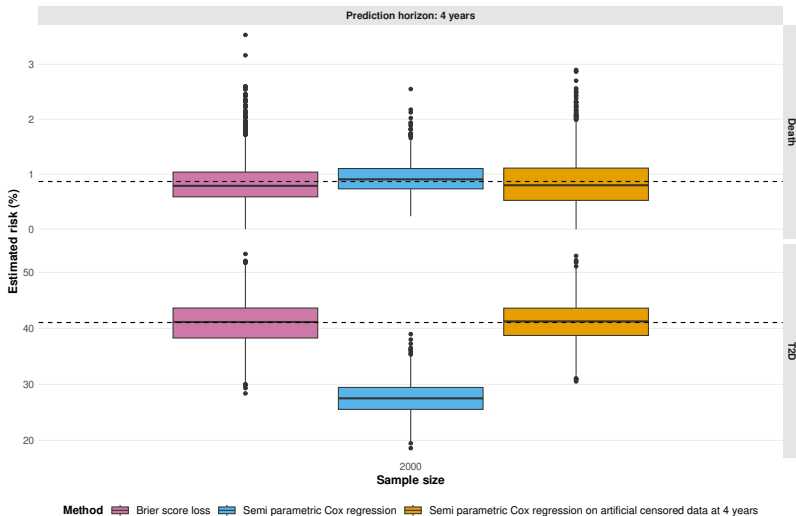
It can gain predictive power

If effects change after four years, an untruncated Cox fit compromises across periods and can learn the wrong effect for the four-year target.

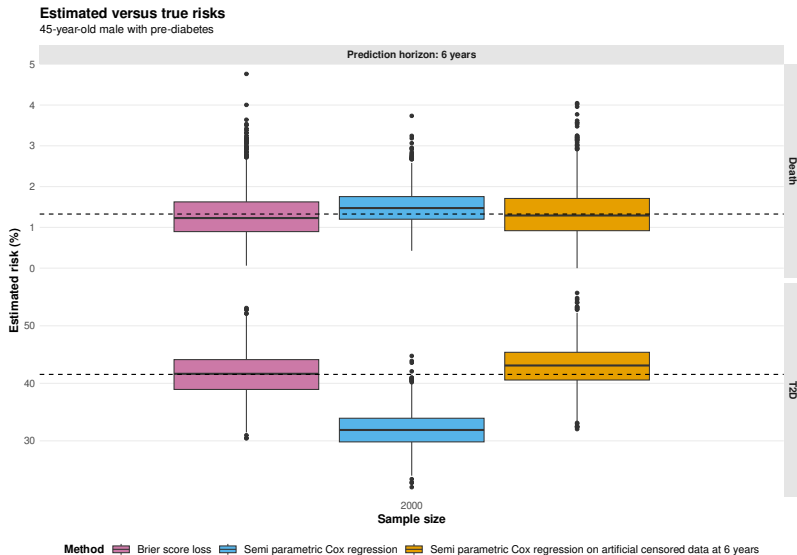
Simulation study: Prediction at 4 years

Estimated versus true risks

45-year-old male with pre-diabetes



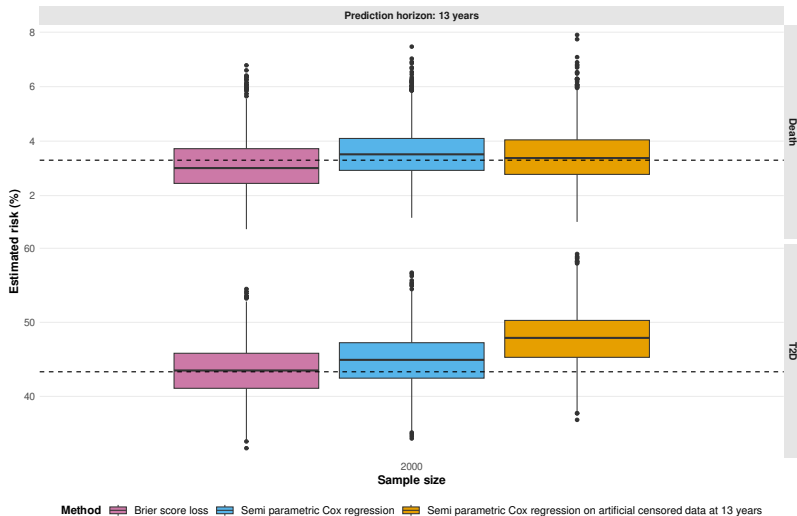
Simulation study: Prediction at 6 years



Simulation study: Prediction at 13 years

Estimated versus true risks

45-year-old male with pre-diabetes



Discussion

- Estimation targets prediction horizon
- The criterion function is non-convex: initialization and repeated starts matter
 - Use partial likelihood and Breslow estimates from a semi-parametric Cox regression model to find start values for hazard ratios and baseline hazard
 - Consider to switch from Brier score to logarithmic loss
- Need to extend framework to illness-death.